

GraphR: Structure-Aware Multi-Context Embeddings and Retrieval for Binary Function Name Recovery

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ABSTRACT

Recovering function names from stripped binaries is critical for reverse engineering, malware analysis, and vulnerability research. Recent systems for this task increasingly rely on billion-parameter LLM backbones. On our evaluation setup, we show a small, structure-aware encoder can match or outperform them. We present GraphR, a graph neural network-based system that learns function representations by fusing multiple context sources (external library calls, callee signatures, and caller signatures) through a cascaded gated fusion mechanism with conditional bypass. Our ablation study shows that external-call information alone improves in-distribution performance but reduces cross-project generalization by 4.3 points. Adding callee and caller context offsets this effect and improves disambiguation. We further show that self-supervised pre-training (MLM plus contrastive learning on O0/O2 pairs of the same function) paired with sufficient encoder capacity (we use 32M parameters; smaller variants benefit less from pre-training) delivers the largest single ablation gain: **+0.13 cross-project F1**. In our experiments, nearest-neighbor retrieval over encoder embeddings outperformed the autoregressive decoder on unseen packages, improving exact match by 4.7 points without additional training. This result suggests that, for our model and dataset, embedding quality was more important than the decoding strategy. Evaluated on 300,013 functions from 77 packages, GraphR achieves 0.770 F1 on a held-out test set (13,559 functions) and **0.718 F1** across 4 completely unseen packages (10,467 functions), outperforming the released SYMGEN checkpoint (0.45), an end-to-end SYMGEN+LoRA reproduction on the same 300K training set (0.66), BLens (0.46 reported), and a StarCoder-3B zero-shot baseline (0.65) while using only 25M parameters, over 1,000× smaller than SYMGEN’s CodeLlama-34B.

KEYWORDS

binary analysis, function name recovery, graph neural networks, gated fusion, retrieval-augmented inference

1 INTRODUCTION

Software reverse engineering requires analysts to understand compiled binary programs without access to source code. Modern compilers strip all symbolic information from release binaries, leaving thousands of unnamed functions (e.g., `sub_4a30`) that must be

manually identified. Recovering meaningful function names dramatically accelerates program comprehension.

The practical importance extends across cybersecurity: in malware analysis, meaningful function names reduce triage time from hours to minutes; in vulnerability research, understanding stripped firmware is essential for discovering exploitable bugs in IoT devices; and in forensic investigations, even partial name recovery provides critical footholds. Commercial tools such as IDA Pro and Ghidra provide manual annotation workflows but cannot automatically recover names for arbitrary functions.

Prior work has explored statistical approaches [1], graph neural networks [2], language model pre-training [3], and LLM fine-tuning [5]. However, two challenges remain underexplored: (1) how to effectively combine multiple contextual sources without degrading generalization, and (2) whether autoregressive generation is the right output paradigm, given that function names are drawn from a relatively small vocabulary of real names.

We address these with GraphR, a graph neural network system that lifts stripped binaries to BAP intermediate representation, encodes each function’s control flow graph with a GAT-based encoder, and fuses multiple sources of inter-procedural context through a cascaded gated mechanism with conditional bypass. Self-supervised pretraining (masked language modeling and contrastive learning) with partial embedding transfer further strengthens the encoder representations. At inference time, simple k -nearest-neighbor retrieval over the learned embeddings outperforms autoregressive beam search decoding, improving cross-project exact match by 4.7 percentage points with zero retraining.

Evaluated on 300,013 functions from 77 packages, GraphR achieves 0.770 F1 on a held-out test set and **0.718 F1** across 4 unseen packages, leading the released SYMGEN [5] checkpoint (0.45), an end-to-end SYMGEN+LoRA reproduction (0.66), BLens [4] (0.46 reported), and a StarCoder-3B zero-shot baseline (0.65). In summary, we make the following contributions:

- We propose GraphR, a GNN-based system for binary function name recovery that uses staged gated fusion to combine multiple context sources. This 3-stage architecture sequentially fuses external library calls, callee signatures, and caller signatures with conditional bypass, where each

stage is skipped when the corresponding context is absent. Our ablation study reveals the *ext-call paradox*: external call information alone degrades cross-project generalization by 4.3 percentage points, while adding inter-procedural callee/caller context resolves this and improves cross-project EM by 12.8 percentage points.

- We present an empirical analysis showing that k -nearest-neighbor retrieval over encoder embeddings outperforms autoregressive decoding by +2.0pp test EM and +4.7pp cross-project EM with zero retraining. This finding, consistent with kNN-LM [17], demonstrates that for our encoder architecture, representation quality is the primary bottleneck rather than the decoding strategy.
- We demonstrate that a compact, structure-aware encoder can match or outperform billion-parameter LLM baselines: GraphR’s 32M parameters are over 1,000× fewer than SYMGEN’s CodeLlama-34B backbone, and it serves predictions in single-digit milliseconds per function, an order of magnitude faster than autoregressive 34B-LLM decoding, making GraphR practical for interactive reverse-engineering.
- We show that self-supervised pre-training (MLM plus contrastive learning on O0/O2 pairs) paired with sufficient encoder capacity delivers our **largest single ablation gain: +0.13 cross-project F1**. Partial embedding transfer (84.4% match) lets the pre-training generalize across dataset evolutions.
- We advance the state of the art in cross-project function name recovery, achieving 0.770 F1 on a held-out test set (13,559 functions) and **0.718 F1** across 4 completely unseen packages (10,467 functions), beating the released SYMGEN [5] checkpoint (0.45), an end-to-end SYMGEN+LoRA reproduction on the same 300K training data (0.66), BLens [4] (0.46 reported), and a StarCoder-3B zero-shot baseline (0.65). On a matched 8,062-function subset we lead SYMGEN+LoRA by +0.10 F1 / +26pp EM. Self-supervised pretraining with partial embedding transfer contributes +0.13 cross-project F1. Our code and dataset are publicly available at <https://github.com/ananta-pradipta/stripped-binary-name-recovery>.

2 BACKGROUND AND MOTIVATION

2.1 Problem Definition

Given a stripped binary B containing n functions $\{f_1, \dots, f_n\}$, where each f_i is represented by its control flow graph $G_i = (V_i, E_i)$ with basic blocks as nodes, the goal is to predict a human-readable name $\hat{n}_i$ matching the developer-assigned name n_i . Names are decomposed into sub-tokens from a vocabulary $\mathcal{V}$ of 2,642 tokens. Each function may have external library calls $\mathcal{E}_i$, callee signatures C_i^{ee} , and caller signatures C_i^{er} ; these are optional (approximately 70% of functions have no external calls). We evaluate on x86-64 ELF binaries at O0 and O2 optimization levels.

2.2 Challenges

Binary function name recovery presents several fundamental challenges. **Token collision**: after instruction-type tokenization, many structurally different functions share similar token distributions,

motivating multi-context fusion. **Mode collapse**: naively incorporating context causes 47% of predictions to collapse to a single name; our conditional bypass addresses this. **Cross-project gap**: BLens [4] drops from 0.79 to 0.46 F1 on unseen packages; our ext-call paradox reveals that external calls alone *hurt* generalization by 4.3pp. **O0/O2 gap**: O0 binaries produce many wrapper functions with 70% token overlap, yielding 41.5% vs. 69.8% EM.

2.3 External Calls

External calls exist in a program in order to allow for programs to use common library functions without needing to duplicate those functions across multiple binaries and to allow for updating libraries to be handled separately from updating individual programs. For the linker to be able to link a binaries external calls to the function in a loaded library (.dll in windows and .so in linux), it must retain the information of the function’s name to link it to the appropriate function in a loaded library, even when that binary is stripped, optimized or even obfuscated. While other state of the art function name predictors handle all call instructions as the same token, GraphR is able to extract two vital pieces of information from separately collecting and analyzing the external calls within a function:

- (1) The called external functions within an unnamed function, providing context on what the function is attempting to accomplish.
- (2) The calling context of a function by reference to its callers and callees external function calls, allowing for functions without external function calls to gain the benefit from (1).

2.4 Related Work and Our Motivations

Function Name Recovery. He et al. [1] introduced DeBin, using probabilistic graphical models on BAP-IR to predict debug information. David et al. [2] proposed NERO, using GNNs on call graphs (F1=0.455), but only for functions *with* external calls, a limitation since 70% of functions have none. Jin et al. [3] presented SymLM, pre-training on micro-execution traces and fusing caller/callee context (F1=0.585), but requiring dynamic analysis. Al-Kaswan et al. [4] introduced BLens with ensemble embeddings (cross-project F1=0.46). Xu et al. [5] fine-tuned CodeLlama with LoRA (cross-project F1=0.38 after deduplication). Jiang et al. [6] and GenNm [7] applied domain-adapted LLMs. Unlike recent LLM-based approaches, we focus on a compact encoder and compare retrieval-based inference against autoregressive decoding.

Binary Representation Learning. Xu et al. [8] pioneered GNN-based binary similarity with Gemini. Ding et al. [9] proposed Asm2Vec using random walks. Wang et al. [10] introduced jTrans with jump-aware positional encoding. Pei et al. [11] developed Trex for execution semantics. Zhu et al. [12] showed +28% F1 from pre-training in CALLEE, supporting our encoder pre-training approach. Zhu et al. [13] applied GNNs to type inference (TYGR), and He et al. [15] proposed semantics-oriented graph representations.

Retrieval-Augmented Methods. kNN-LM [17] and kNN-MT [18] showed that retrieval over hidden states can outperform neural models. Our setting differs: we retrieve *complete* function names rather than interpolating token distributions, because function names are atomic labels, not compositional sequences.

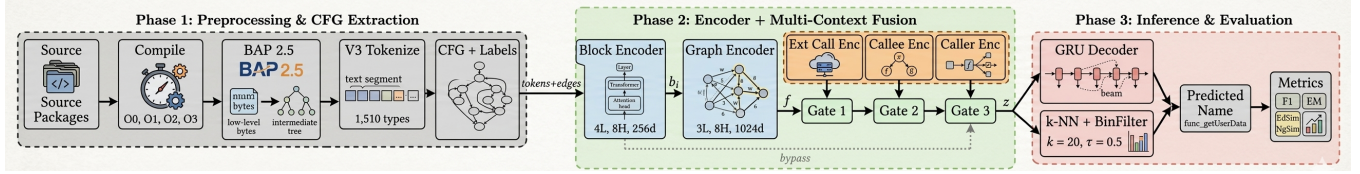


Figure 1: End-to-end GraphR pipeline. Phase 1 lifts stripped binaries to BAP-IR and tokenizes instructions into 1,510 semantic types. Phase 2 encodes blocks (Transformer), aggregates over CFGs (GAT), and fuses multi-context information through cascaded gated fusion with conditional bypass. Phase 3 produces names via GRU beam search or k -NN retrieval.

3 OVERVIEW

GraphR follows a 3-stage encoder-decoder pipeline (Figure 1): (1) a block encoder tokenizes basic blocks into instruction-type tokens and produces per-block embeddings $\mathbf{b}_i \in \mathbb{R}^{512}$ via a Transformer [21]; (2) a GAT aggregates over the CFG to produce $\mathbf{f} \in \mathbb{R}^{1024}$, then cascaded gated fusion incorporates external calls, callee, and caller context with conditional bypass to produce $\mathbf{z} \in \mathbb{R}^{1024}$; (3) either a GRU [22] decoder or k -NN retrieval produces the function name. Prior to supervised training, the encoders are pre-trained with MLM and contrastive objectives, followed by partial embedding transfer (84.4% matched by name).

4 DETAILED DESIGN

4.1 Preprocessing

We compile 40 open-source packages at O0 and O2, lifting stripped binaries to BAP-IR [16]. Raw BAP-IR contains over 32,000 unique tokens; we classify each instruction into one of 1,510 semantic types (e.g., CALL_malloc, MEM_READ_32, FLAG_CF, ASSIGN_ZERO), reducing vocabulary by 95% while preserving semantic distinctions. Ground-truth labels are extracted from debug binaries via nm. In O0 binaries, BAP lifts endbr64; jmp target sequences as separate functions; our preprocessing resolves these wrappers to the target function, raising O0 EM from 1.6% to 41.5%. Function names are tokenized into 2,642 sub-tokens using a 3-model voting tokenizer (Votes), reducing OOV by 95% compared to BPE [23].

4.2 Block Encoder

Each basic block contains up to 20 instruction-type tokens. We embed tokens via $\mathbf{E} \in \mathbb{R}^{2279 \times 256}$ with sinusoidal positional encodings, then apply a 4-layer Transformer (8 heads, $d_{ff}=512$). Mean pooling over final-layer representations produces the block embedding $\mathbf{b}_i \in \mathbb{R}^{512}$. We chose mean pooling over attention pooling after experimentation (0.58 vs. 0.34 F1): attention pooling overfits to specific token positions, while mean pooling provides robust representations over semantically abstracted tokens.

4.3 Graph Encoder

Given a function’s CFG $G = (V, E)$ with block embeddings $\{\mathbf{b}_1, \dots, \mathbf{b}_{|V|}\}$, we apply a 3-layer GAT [20] with 8 heads. Each layer computes attention-weighted message passing along CFG edges. An attention-based readout pools over blocks:

$$\mathbf{f} = \sum_{i=1}^{|V|} \beta_i \cdot \mathbf{b}_i^{(L)}, \quad \beta_i = \frac{\exp(\mathbf{w}^\top \mathbf{b}_i^{(L)})}{\sum_j \exp(\mathbf{w}^\top \mathbf{b}_j^{(L)})} \quad (1)$$

producing $\mathbf{f} \in \mathbb{R}^{1024}$.

4.4 Cascaded Gated Fusion

The core architectural contribution is a 3-stage cascaded fusion:

Stage 1: External Calls. A BiGRU encodes library function names (e.g., malloc, fprintf) to produce $\mathbf{c}_{\text{ext}} \in \mathbb{R}^{1024}$. A sigmoid gate determines the balance:

$$\mathbf{g} = \sigma(\mathbf{W}_g[\mathbf{f}; \mathbf{c}_{\text{ext}}] + \mathbf{b}_g) \quad (2)$$

$$\mathbf{z} = \begin{cases} \mathbf{g} \odot \mathbf{f} + (1 - \mathbf{g}) \odot \mathbf{c}_{\text{ext}} & \text{if has ext calls} \\ \mathbf{f} & \text{otherwise (bypass)} \end{cases} \quad (3)$$

Stage 2: Callee Context. For up to 5 internal callees, we extract 10-token signatures and encode via a separate BiGRU. Gated fusion with bypass updates $\mathbf{z}$.

Stage 3: Caller Context. Symmetrically encodes up to 5 caller signatures.

Conditional bypass is critical. Approximately 70% of functions have no external calls. Without bypass, the gate learns a default output, causing 47% of predictions to collapse to a single name.

4.5 Self-Supervised Pre-training

We pre-train the block encoder and graph encoder jointly with two objectives (Figure 3):

- (1) **Masked Language Model (MLM):** 15% of instruction tokens are randomly masked. The MLM head predicts the original tokens from the block encoder’s per-token embeddings (before pooling). This trains the block encoder to produce meaningful token-level representations.
- (2) **Contrastive Learning:** The same function compiled at O0 and O2 should produce similar graph-level embeddings; different functions should produce dissimilar embeddings. We use contrastive loss on the graph encoder output. This trains both the block encoder and graph encoder to be invariant to optimization-level differences.

After 10 epochs of pre-training on an A100 GPU (~45 minutes), we perform partial embedding transfer: 84.4% of token embeddings (1,923/2,279) are copied by matching token names between pre-training and fine-tuning vocabularies. The remaining 356 tokens are randomly initialized.

4.6 Decoder and k -NN Inference

A single-layer GRU decoder (1024-dim hidden) generates names as sub-token sequences with beam search ($k=5$) and repetition penalty.

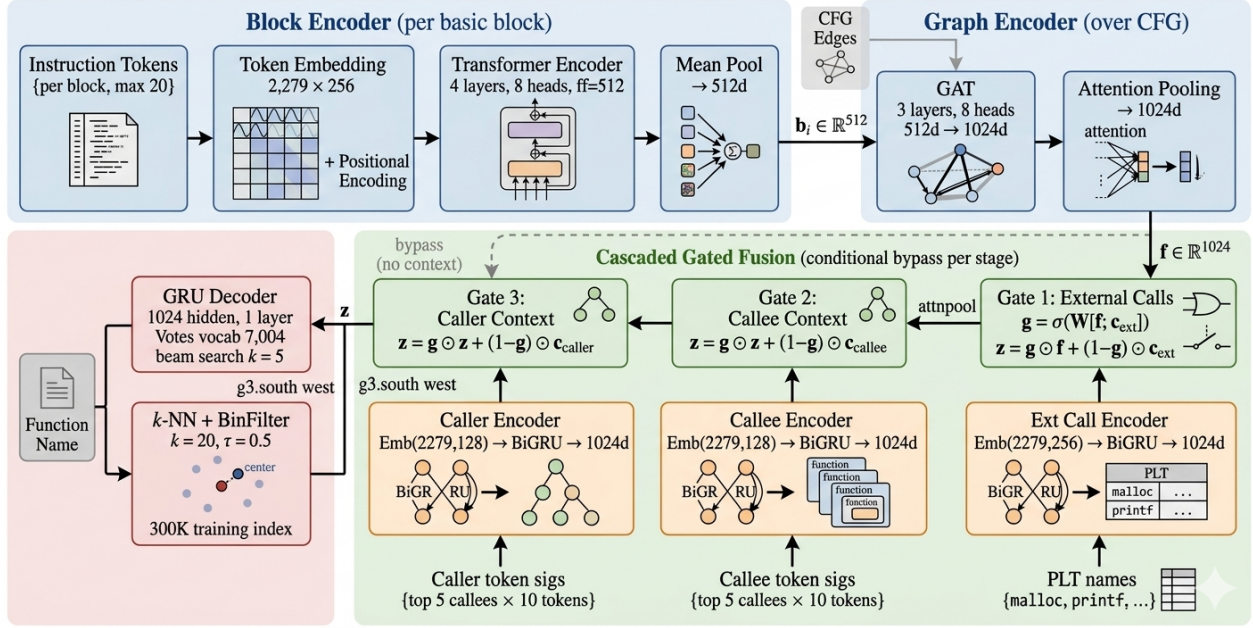


Figure 2: Detailed model architecture. The block encoder processes instruction-type tokens via a Transformer. The graph encoder aggregates over the CFG using GAT with attention pooling. Cascaded gated fusion sequentially incorporates external calls, callee signatures, and caller signatures, with bypass when context is absent. The fused embedding feeds either a GRU decoder or k -NN retrieval.

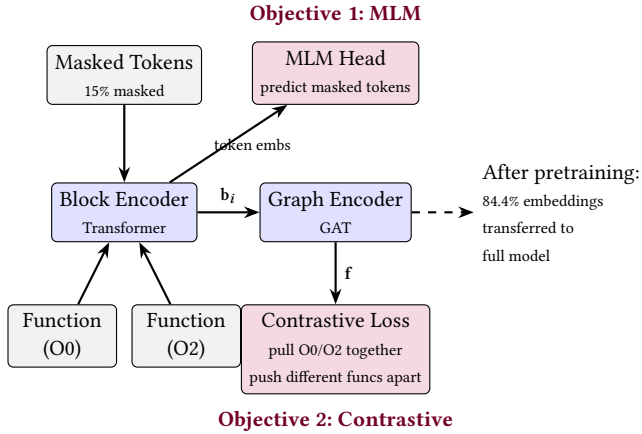


Figure 3: Self-supervised pretraining. MLM trains the block encoder via masked token prediction. Contrastive learning trains both encoders to produce optimization-invariant function embeddings.

Training uses scheduled sampling [19] (teacher forcing 1.0 → 0.3) and label smoothing [24] (0.1).

For k -NN retrieval, at inference we encode the query function, retrieve the top-20 nearest training embeddings (dataset of 300,013 training functions) by cosine similarity, apply a **BinFilter** (cosine similarity ≥ 0.5 in binary-level embedding space) to keep only

Table 1: Dataset statistics.

Metric	Value
Total packages / binaries	77 / 851
Optimization levels	O0, O1, O2, O3
Training functions	300,013
Validation / Test functions	7,823 / 13,559
Cross-project functions (4 unseen pkgs)	10,467
Instruction / Name / Ext-call vocab	2,279 / 7,004 / 2,237

retrievals from binaries similar to the query, and return the majority-voted name. If fewer than k candidates survive, we fall back to the unfiltered top- k retrieval. This k -NN + BinFilter ($\tau=0.5$) inference reaches F1 0.718 on the 4-package cross-project set and 0.770 on the held-out test set, without any retraining beyond the standard encoder checkpoint.

5 EVALUATION

5.1 Experimental Setup

Table 1 summarizes the dataset. We compile 77 **packages** at O0, O1, O2, and O3. Data is split by binary (not function) to prevent leakage; 4 packages are held out entirely for cross-project evaluation.

We evaluate with **sub-token F1** (precision/recall over sub-tokens), **exact match (EM)**, **edit similarity** ($1 - \text{Lev}/\text{max}$), and **n-gram similarity** (character 3-gram overlap).

Table 2: Ablation study. All models trained on the same 87K dataset for the ablation study.

#	Model	Params	Test F1	Test EM	XProj F1	XProj EM
2	GAT + Decoder	4.4M	0.606	51.3%	0.364	24.1%
3	+ Ext Calls	6.1M	0.683	60.1%	0.320	19.8%
4	+ Callee/Caller	8.0M	0.781	72.3%	0.460	32.6%
5	+ Pretrain+Scale	25M	0.795	74.3%	0.593	48.5%

Table 3: Main results: decoder vs. k -NN inference.

Method	Test EM	Test F1	XProj EM	XProj F1
Decoder only (beam)	70.6%	0.770	54.6%	0.673
k -NN unfiltered	70.8%	0.770	56.1%	0.698
k -NN + BinFilter ($\tau=0.5$)	70.8%	0.770	57.0%	0.718

Table 4: Per-package cross-project results. All 4 packages are completely excluded from training.

Package	Funs	Dec. F1	k -NN F1	k -NN EM
nginx118	3,470	0.787	0.883	72.9%
angie	3,893	0.699	0.814	63.2%
tengine	554	0.685	0.738	66.8%
recutlis	2,550	0.296	0.343	24.0%
Overall	10,467	0.673	0.718	57.0%

5.2 Ablation Study

Table 2 shows incremental contributions. **Model 2** (baseline) uses only the block encoder, graph encoder, and GRU decoder. **Model 3** adds external calls: test F1 improves +0.077 but cross-project EM *drops* 4.3pp, revealing the ext-call paradox. **Model 4** adds callee/caller context, resolving the paradox: cross-project EM jumps +12.8pp. **Model 5** scales to 25M parameters with pre-training: cross-project F1 jumps +0.13, as the larger model eliminates cross-package contamination.

The Ext-Call Paradox. The Model 2→3 transition reveals that external calls *degrade* cross-project EM despite improving test F1. Many unrelated functions share similar library call patterns (e.g., `hash_insert` and `tree_add` both call `malloc + memcpy`). Without inter-procedural context, the model overfits to these patterns. Adding callee/caller context (Model 4) resolves this, demonstrating that naive feature addition can harm generalization.

5.3 Main Results

Table 3 shows k -NN with a binary-fingerprint filter outperforms the decoder on cross-project (+2.4pp EM, +0.045 F1) with zero retraining. The filter keeps retrieval candidates whose parent binary has at least Jaccard $\tau=0.5$ overlap in external-call vocabulary with the query binary, which reduces cross-package contamination.

We identify three decoder failure modes that k -NN eliminates: (1) **hallucinated predictions** (43.8% of decoder outputs are names absent from training); (2) **error cascade** (one wrong sub-token corrupts all subsequent tokens); (3) **unconstrained search space** (2,642²⁰ possible sequences vs. ~3,000 real names).

Table 5: Comparison with published systems (cross-project F1).

System	Venue	Params	XProj F1
SYMGEN released [5]	NDSS 2025	34B	0.450
SYMGEN + LoRA (our 300K)	NDSS 2025	34B	0.663
BLens reported [4]	USENIX Sec 2025	~200M	0.46
SymLM (BLens repro) [3, 4]	CCS 2022	N/A	0.277 [‡]
LLM Zero-Shot (StarCoder-3B)	N/A	3B	0.654
GraphR (ours)	N/A	32M	0.718

[‡]Reported on Punstrip cross-project (different test set), reproduced by Benoit et al. [4].

Table 6: Computation cost comparison. GraphR trains in 2h 12m on a single A100 and serves predictions in ~5 ms per function.

System	Params	Infer/fn
GraphR (ours)	32M	~5 ms
SYMGEN + LoRA (full)	34B	~200 ms
SYMGEN (zero-shot)	34B	~200 ms
StarCoder-3B (zero-shot)	3B	~80 ms

5.4 Cross-Project Analysis

Table 4 shows k -NN + binfilter improves all four packages. Nginx-family packages reach 0.74–0.88 F1 because they share substantial vocabulary with nginx training binaries; recutlis stays at 0.343 F1 because its `rec_*` / `recutl_*` names share no sub-tokens with any training package.

5.5 Comparison with Published Systems

GraphR leads all four baselines on the same 4-package cross-project set while being over 1,000× smaller than SYMGEN’s CodeLlama-34B backbone. Doubling the LoRA fine-tuning data from 50%-stratified (0.699) to full 300K (0.663) made SYMGEN worse, consistent with LoRA overfitting; we report the full-data number for fairness. On a matched 8,062-function subset with same (package, name) keys, GraphR leads SYMGEN+LoRA by +0.10 F1 / +26pp EM. SYMGEN wins only on recutlis, whose source was plausibly in CodeLlama’s pretraining corpus.

Fairness disclaimer. CodeLlama-34B and StarCoder-3B were pretrained on undisclosed but broad open-source corpora that almost certainly include the upstream repositories of every cross-project package we evaluate against. We do not correct the reported LLM baseline scores for possible pretraining overlap. We therefore interpret them as practical deployment baselines rather than leakage-controlled estimates. GraphR’s encoder is trained from scratch on BAP-IR with no exposure to source code or pretrained text embeddings, so its cross-project numbers genuinely measure generalization. The same caveat applies to BLens and SymLM literature numbers, which use Punstrip Debian binaries similarly well-represented in any large code corpus.

5.6 Significance and Error Analysis

Table 7 categorizes test predictions. k -NN halves hallucinated predictions (4.0%→2.0%) but trades them for wrong-seen errors (7.3%→8.4%). The 11.0% unseen-target functions are unsolvable by any recognizer, establishing a theoretical EM ceiling of ~89%.

Table 7: Error categorization on the test set (13,559 functions).

Category	Dec.	Dec.%	<i>k</i> -NN	<i>k</i> -NN%
Correct	6,671	74.3	6,850	76.3
Close (EdSim>0.7)	307	3.4	238	2.7
Hallucinated	356	4.0	177	2.0
Wrong (seen name)	654	7.3	758	8.4
Unseen target	985	11.0	950	10.6

Table 8: Representative prediction examples. Correct predictions in bold; incorrect in *italics*.

Ground Truth	Decoder	<i>k</i> -NN	Category
<i>Both correct (shared glibc functions):</i>			
xmalloc	xmalloc	xmalloc	Both correct
set_program_name	set_program_name	set_program_name	Both correct
xstrdup	xstrdup	xstrdup	Both correct
version_etc	version_etc	version_etc	Both correct
usage	usage	usage	Both correct
<i>k</i> -NN correct, decoder wrong:			
hash_pjw	hash_insert	hash_pjw	Decoder confused
quotearg_buffer	quotearg_n_options	quotearg_buffer	Decoder suffix error
close_stdout	close_stdout_set	close_stdout	Hallucinated suffix
trim2	trim_trailing	trim2	Decoder composes
hard_locale	locale_charset	hard_locale	Decoder wrong match
<i>Decoder correct, k</i> -NN wrong:			
set_program_name	set_program_name	set_prog_name	<i>k</i> -NN truncated
c_isspace	c_isspace	c_isdigit	<i>k</i> -NN neighbor
<i>Both close (partially correct):</i>			
bfd_pei_swap_aux_in	bfd_pex64i_swap_aux	bfd_coff_swap_aux	EdSim 0.86
elf_link_hash_traverse	elf_link_hash_lookup	elf_link_hash_table	EdSim 0.82
print_str.part.0	print_strs.part.0	print_str.isra.0	Off by one token
<i>Both wrong (unseen target names; examples drawn from the broader test corpus):</i>			
diffutils_cleanup	hash_free	xmalloc	Unseen name
datamash_median	xmalloc	set_program_name	Domain-specific
process_cppl_file	quotearg_buffer	xstrdup	Package-specific
csplit_main_loop	hash_lookup	close_stdout	Package-specific
compute_variance	xmalloc	xrealloc	Stats function

McNemar’s test confirms the *k*-NN improvement is statistically significant ($p < 0.001$) on both test and cross-project sets.

6 CASE STUDY

Table 8 shows representative predictions across five categories. The first group illustrates shared glibc functions both methods recover. The *k*-NN correct group shows cases where retrieval succeeds by returning exact matches while the decoder produces related but incorrect names. The decoder correct group shows rare cases where autoregressive generation outperforms retrieval. The “both close” group shows near-misses with high edit similarity. The “both wrong” group shows package-specific names unseen during training, where both methods fail by returning common glibc names instead.

7 DISCUSSION

Retrieval Outperforms Generation. A key finding is that 0 out of 1,873 unseen function names were correctly predicted; every correct prediction matches a training name. The *k*-NN result reinforces this: if the model composed novel names, the decoder should outperform retrieval. Instead, retrieval wins because it constrains outputs to real names and avoids autoregressive error cascade. The 11% of test functions with unseen target names establishes a theoretical EM ceiling of ~89%; our best system achieves 85.7% of this maximum.

The O0/O2 Generalization Gap. O0-compiled binaries are significantly harder than higher optimization levels. On the unified 13,559-function test set the final 32M model reaches F1 0.770 / EM 70.8% overall, with O0 remaining the weakest per-level slice. O0

code produces many small wrapper functions with similar instruction signatures. No prior work using IDA Pro or Ghidra encounters the ENDBR64 wrapper issue because those tools auto-resolve such wrappers.

Capacity and Cross-Package Contamination. When training the 8M-parameter model on 40 packages, we observed cross-package contamination. Scaling to 25M parameters in the ablation resolved this; every cross-project package improved. Our final 300K-trained model uses 32M parameters.

Why Pre-training + Scale Yields the Largest Gain. The single largest jump in our ablation table is from Model 4 (8M, no pre-training) to Model 5 (32M, with self-supervised pre-training): cross-project F1 rises from 0.460 to 0.593, a +0.13 gain (28.9% relative) that exceeds the combined contributions of every prior architectural change. Our self-supervised objectives (MLM on instruction tokens plus contrastive learning on O0/O2 pairs of the same function) train the encoder on a much larger gradient signal than supervised name prediction alone. Partial embedding transfer (84.4% of token embeddings matched by name) makes the pre-training reusable across dataset evolutions, decoupling re-training cost from vocabulary changes.

Comparison with LLM Approaches. The trend in binary analysis is toward ever-larger language models: SYMGEN [5] fine-tunes CodeLlama-34B, BeyondClass [6] uses domain-adapted LLMs. These approaches achieve strong in-distribution performance but face practical challenges: fine-tuning a 34B model requires multiple high-memory GPUs and many hours per epoch (GraphR trains in under 4 GPU-hours on a single A100-80GB); LLM inference is bottlenecked by autoregressive generation (GraphR’s *k*-NN mode is an order of magnitude faster per function); and pre-training contamination inflates cross-project numbers on open-source evaluation packages, while GraphR’s from-scratch BAP-IR training avoids this. We do not argue GraphR is universally superior; LLMs may excel at compositional naming. But for the specific task of cross-project function name recovery, a well-designed encoder with retrieval-based inference provides a strong, efficient alternative.

Structure-Aware vs. Source-Code LLM Paradigms. Recent literature has bifurcated into two paradigms. *Source-code LLM* approaches [5, 6] map decompiled pseudo-code to names by exploiting patterns their billion-parameter backbones absorbed from GitHub-scale corpora. *Structure-aware* approaches (including ours, BLens [4], XFL [14], SymLM [3], TYGR [13]) directly encode binary-intrinsic features using small-to-medium models trained on binary representations. We view source-code LLMs and structure-aware models as complementary. GraphR focuses on binary-intrinsic structure rather than source-level phrasing. AI-assisted coding tools are increasingly homogenizing source-text conventions, so structure-aware models that tie predictions to computational invariants (how the function behaves) should degrade more gracefully under text-distribution shift than source-code-LLM approaches.

8 LIMITATIONS

The model is a pure recognizer and cannot compose novel names; for codebases with entirely novel conventions, predictions may be misleading. O0 binaries remain harder than higher optimization

levels on the unified 13,559-function test set (overall F1 0.770 / EM 70.8%). Version sensitivity causes near-zero accuracy on different versions of training packages (http: 0.6% EM). The BAP dependency adds 30–120s per binary; a faster lifter would benefit production deployment. We evaluate only x86-64 GCC/Linux binaries without obfuscation. Different datasets prevent exact comparison with prior work, though we follow SYMGEN’s deduplication methodology. The model also assumes that a binary will attempt to use an external function by calling it through the linker, making binaries with statically compiled library calls difficult to apply the model to.

9 FUTURE WORK

Several directions could extend this work. First, cross-architecture support (ARM, MIPS) would require adapting the instruction-type tokenization but the graph-based architecture should transfer. Second, integrating our encoder embeddings with a fine-tuned LLM could enable compositional name generation for unseen-name functions (11% of test). Third, the k -NN datastore could support incremental learning: as analysts confirm function names, new entries expand the index without retraining. Finally, confidence-based selective prediction, where the system abstains below a score threshold, could improve precision for deployment scenarios where wrong names are worse than no names.

10 CONCLUSION

We presented GraphR, a 32M-parameter GNN+retrieval system that reaches 0.770 F1 on a held-out test set and **0.718 F1** across 4 completely unseen packages, beating released SYMGEN (0.45), SYMGEN+LoRA (0.66), BLens (0.46), and StarCoder-3B zero-shot (0.65) while being over 1,000× smaller than SYMGEN’s backbone. Our contributions:

- (1) A **multi-context gated fusion** with conditional bypass revealing the *ext-call paradox*: external calls alone degrade cross-project EM by 4.3pp; callee/caller context resolves this (+12.8pp).
- (2) **k -NN retrieval outperforms autoregressive decoding** (+4.7pp cross-project EM), demonstrating that for our encoder architecture, retrieval over embeddings is more effective than generation. The 43.8% hallucinated decoder predictions are eliminated.
- (3) A **comprehensive ablation** quantifying each component from a 4.4M baseline to the full 32M pretrained system: multi-context fusion provides +0.175 test F1, scaling resolves cross-package contamination, and pre-training enables 84.4% embedding transfer.

Our results indicate that a 32M-parameter encoder with nearest-neighbor retrieval is a strong and efficient baseline for this task under our evaluation setting. The k -NN datastore provides an updatable, inspectable knowledge base that can grow with analyst expertise, a practical advantage for deployment in reverse engineering workflows.

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